BreakBite

Complete Project Documentation and Rebuild Guide

Made with Napkin



The Problem: Canteen Congestion

University students often waste their limited 15–45 minute breaks in long queues.

Students waste

Students waste part of the break waiting.

Often cannot

Often cannot decide what to eat in time.

Target user

A first-year student with 30 minutes and limited money who wants a fast, clear decision.

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Three Ideas to Save Time

BreakBite solves the canteen rush with pre-ordering, real-time data, and smart planning.

Pre-order

Choose food from a live menu, get a token, and collect when ready.

Rush Radar

Show how busy the canteen is so students know when to come.

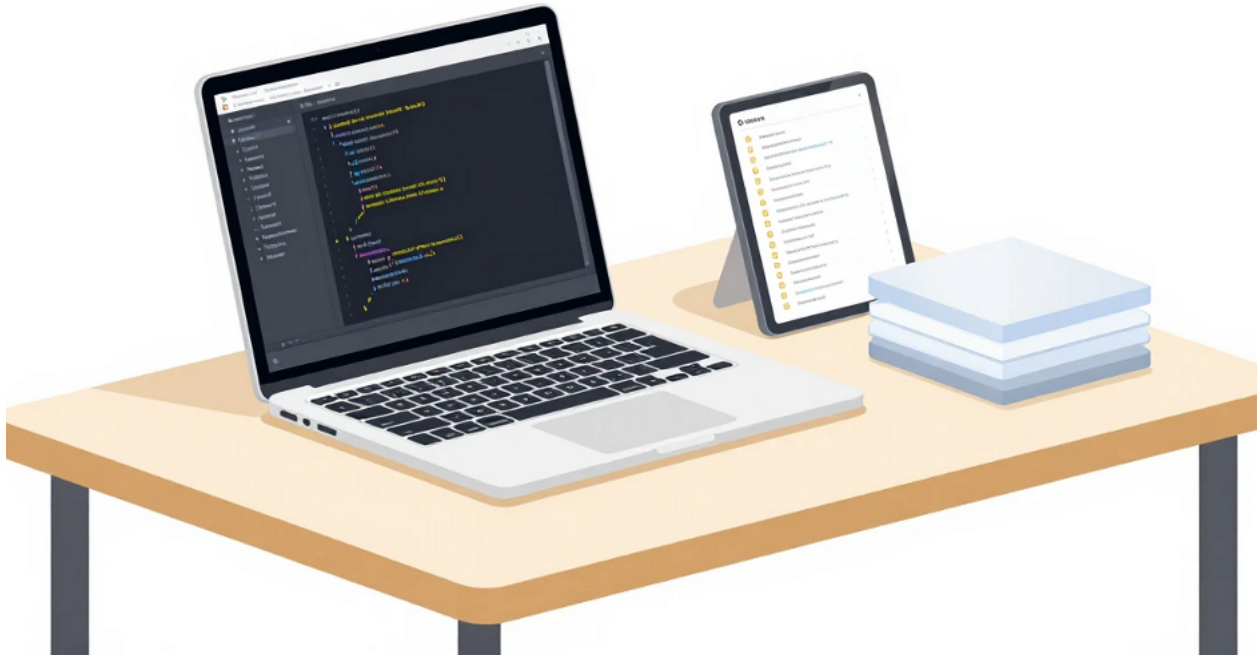
Break Planner

Enter break time and money to get a suggested set of items that fits both.

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Architecture and Technology

A lightweight, real-time stack designed for rapid deployment and collaboration.



01

Frontend

Plain HTML5, CSS3, and vanilla JavaScript (ES modules) for zero-build simplicity.

02

Database

Cloud Firestore for live updates (onSnapshot) between student phones and staff tablets.

03

Authentication

Firebase Authentication for secure staff login.

04

Hosting

GitHub Pages for easy access and version control.

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The Smart Break Planner

Solving the knapsack problem to maximize student satisfaction within time and budget limits.

0/1 and unbounded

Solves a version of the 0/1 and unbounded knapsack problem.

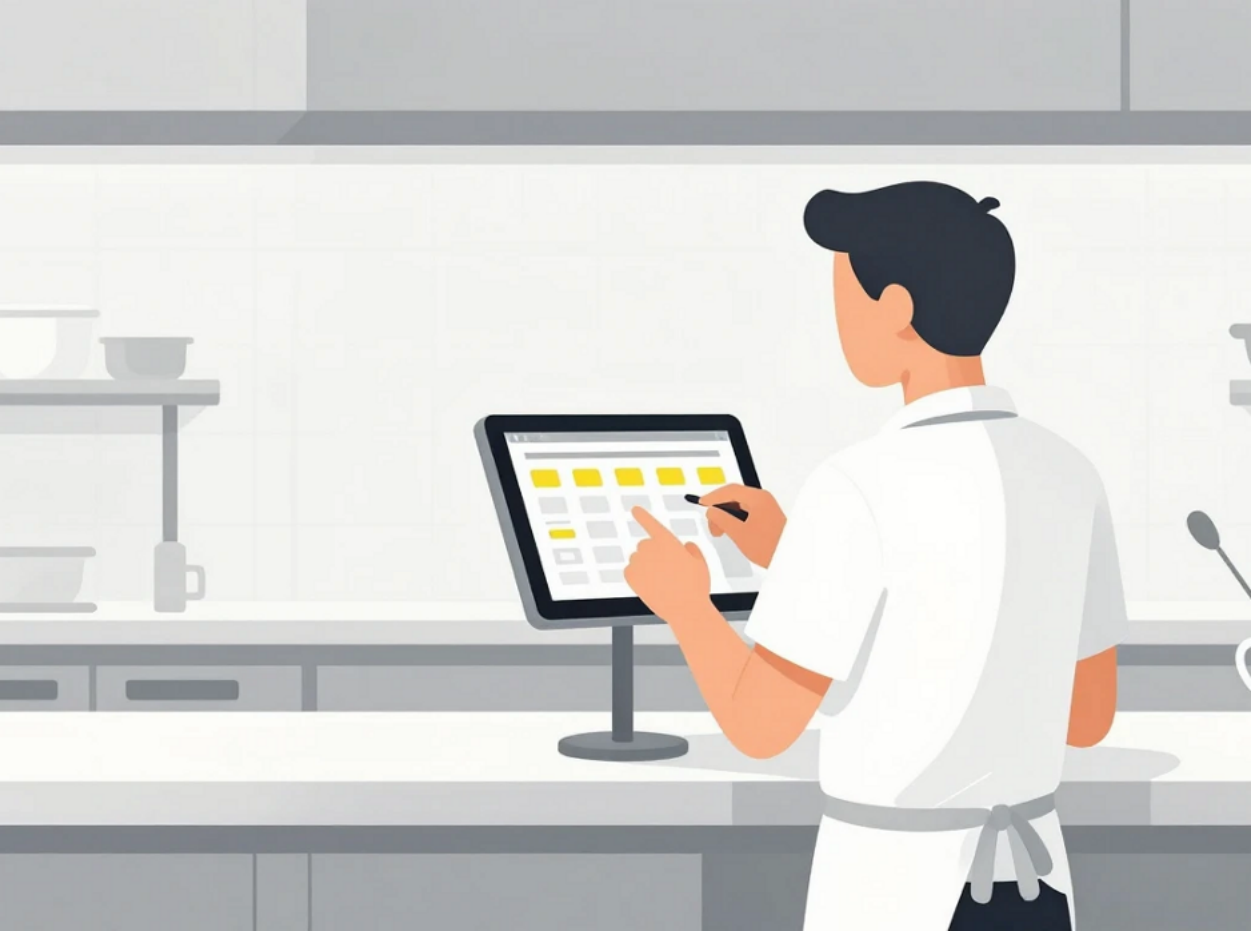
Dynamic programming

Dynamic programming table calculates the best answer quickly for small menus.

Time and price

Considers both time (minutes to eat) and price (Tk) against student limits.

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Efficient Canteen Management

The staff dashboard provides total control over the canteen's daily operations.

Live order board

Live order board with status advance buttons (Received, Preparing, Ready).

Menu control

Menu control: Add, edit, or mark items as sold out.

Food Waste Saver

Food Waste Saver: Apply discounts from 5% to 90% to clear stock.

End of Day

End of Day: Automated summary reports and token reset.

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Future Work and Roadmap

Expanding BreakBite with sustainability and variety in mind.



01

Green Token

Student-side feature for reusable containers.

02

Planner Variety

Rules to prevent repeating the same cheap item too often.

03

Security

Tightening Firestore rules for production-level data safety.

04

Automation

Scheduled Cloud Functions for daily resets.

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